SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 3: 3. Water Jug Problem

Aim: To measure 2 litres using a 4-litre jug and a 3-litre jug.

Python Program

```
from collections import deque

def valid_state(state):
    a, b = state
    return 0 <= a <= 4 and 0 <= b <= 3

def water_jug():
    start = (0, 0)
    q = deque([(start, [start])])
    visited = set()

    while q:
        (a, b), path = q.popleft()
        if (a, b) in visited:
            continue
        visited.add((a, b))

        if a == 2 or b == 2:
            return path

        transfer_ab = min(a, 3 - b)
        transfer_ba = min(b, 4 - a)
        next_states = [
            (4, b), (a, 3), (0, b), (a, 0),
            (a - transfer_ab, b + transfer_ab),
            (a + transfer_ba, b - transfer_ba)
        ]
        for state in next_states:
            if valid_state(state) and state not in visited:
                q.append((state, path + [state]))
    return None

for state in water_jug():
    print(state)
```

Sample Output

```
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
```

Result: The program demonstrates the stated Artificial Intelligence problem.